

# CAP 6. CAMADA DE REDE

## AULA 6: ICMP E TRACEROUTE

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INE5422 REDES DE COMPUTADORES II

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# ICMP: Internet Control Message Protocol

Usado por hosts, routers, para comunicações a nível de rede

- Relato de erros: protocolo, porta, rede, host não encontrado
- echo request/reply (usado pelo ping)

## Msgs ICMP

- são transportados em pacotes IP

Mensagens ICMP: type, code mais primeiros 8 bytes do datagrama IP causando erro

| Type | Code | description                                   |
|------|------|-----------------------------------------------|
| 0    | 0    | echo reply (ping)                             |
| 3    | 0    | dest. network unreachable                     |
| 3    | 1    | dest host unreachable                         |
| 3    | 2    | dest protocol unreachable                     |
| 3    | 3    | dest port unreachable                         |
| 3    | 6    | dest network unknown                          |
| 3    | 7    | dest host unknown                             |
| 4    | 0    | source quench (congestion control - not used) |
| 8    | 0    | echo request (ping)                           |
| 9    | 0    | route advertisement                           |
| 10   | 0    | router discovery                              |
| 11   | 0    | TTL expired                                   |
| 12   | 0    | bad IP header                                 |

# ICMP: Internet Control Message Protocol

## Msgs ICMP

- são transportados em pacotes IP

## Mensagens ICMP:

- type, code mais primeiros 8 bytes do datagrama IP causando erro

The image shows a Wireshark packet capture window titled '\*Ethernet'. The packet list shows two packets: packet 77 is an ICMP Echo (ping) request from 150.162.59.211 to 150.162.2.10, and packet 78 is the corresponding Echo (ping) reply from 150.162.2.10 to 150.162.59.211. The packet details pane for packet 78 is expanded, showing the following information:

- Frame 81: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0
- Ethernet II, Src: IntelCor\_01:3b:02 (00:27:0e:01:3b:02), Dst: IETF-VRRP-VRID\_42 (00:00:5e:00:02:0a)
- Internet Protocol Version 4, Src: 150.162.59.211, Dst: 150.162.2.10
- Internet Control Message Protocol
  - Type: 8 (Echo (ping) request)
  - Code: 0
  - Checksum: 0x4d35 [correct]
  - [Checksum Status: Good]
  - Identifier (BE): 1 (0x0001)
  - Identifier (LE): 256 (0x0100)
  - Sequence number (BE): 38 (0x0026)
  - Sequence number (LE): 9728 (0x2600)
  - [Response frame: 82]
- Data (32 bytes)
  - Data: 6162636465666768696a6b6c6d6e6f707172737475767761...

The packet bytes pane shows the raw data of the ICMP Echo request. The first 8 bytes (00 00 5e 00 01 42 00 27) correspond to the destination MAC address. The next 8 bytes (0e 01 3b 02 08 00 45 00) correspond to the source MAC address. The next 20 bytes (02 0a 08 00 4d 35 00 01 00 26 61 62 63 64 65 66 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76) correspond to the ICMP Echo request data, which includes the identifier and sequence number. The last 8 bytes (77 61 62 63 64 65 66 67) correspond to the data payload.

| No. | Time      | Source         | Destination    | Protocol | Length | Info                                |
|-----|-----------|----------------|----------------|----------|--------|-------------------------------------|
| 77  | 9.9664... | 150.162.59.211 | 150.162.2.10   | ICMP     | 74     | Echo (ping) request id=0x0001, seq= |
| 78  | 9.9666... | 150.162.2.10   | 150.162.59.211 | ICMP     | 74     | Echo (ping) reply id=0x0001, seq=   |

Internet Control Message Protocol (icmp), 40 bytes

Packets: 124 · Displayed: 124 (100.0%) · Dropped: 0 (0.0%) | Profile: Default

# Traceroute e ICMP

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O transmissor envia uma série de segmentos UDP para o destino (ICMP Echo Request)

- 1o possui TTL = 1, 2o possui TTL = 2 ... Incrementa TTL em cada tentativa até encontrar o destino

Quando o enésimo datagrama chega ao enésimo roteador:

- O roteador descarta o datagrama
- E envia à origem uma mensagem ICMP (type 11, code 0)
- A mensagem inclui o nome do roteador e o endereço IP

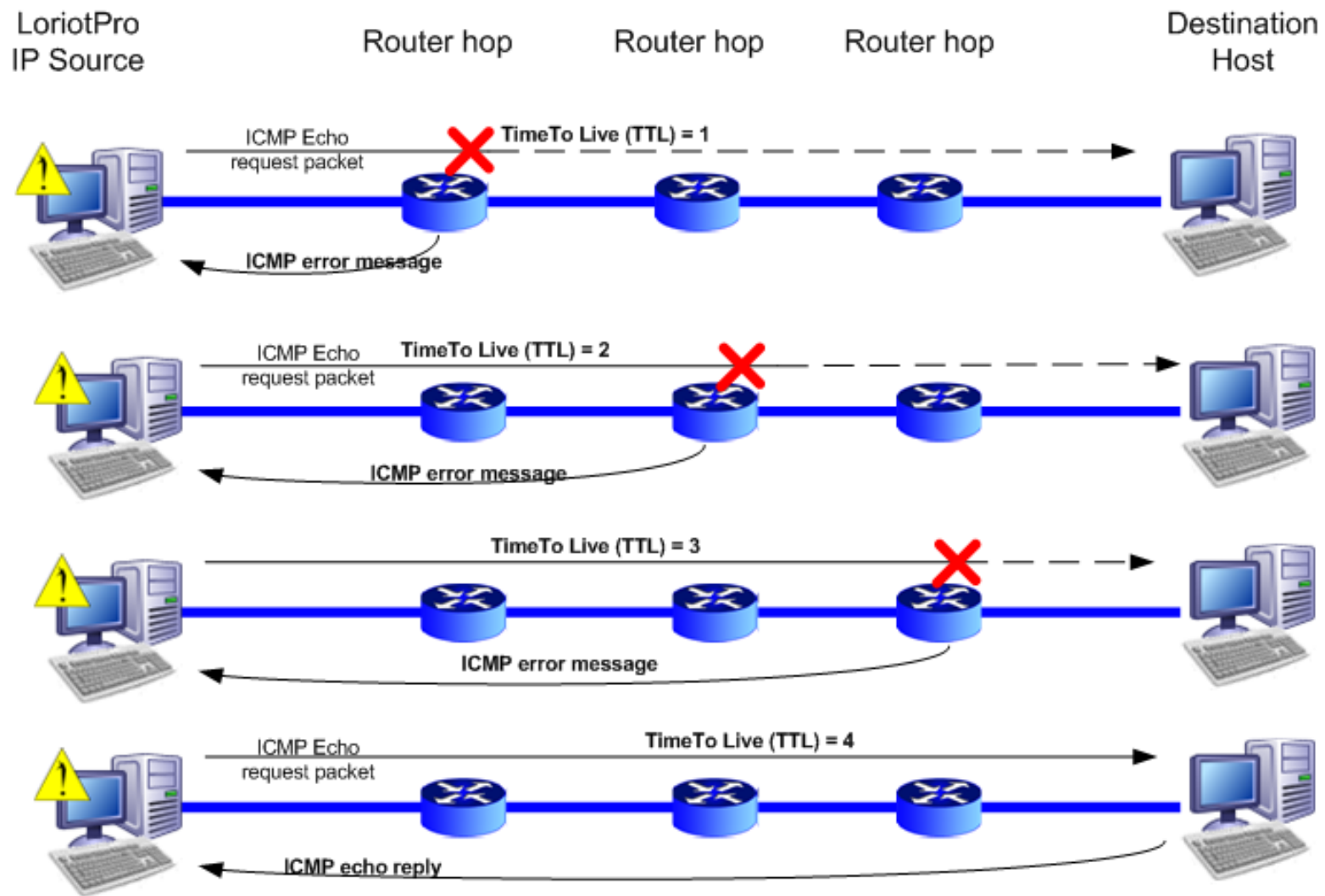
Quando a mensagem ICMP chega, a origem calcula o RTT

- O traceroute faz isso três vezes

Critério de interrupção

- O segmento UDP finalmente chega ao hospedeiro de destino
- O destino retorna o pacote ICMP “Echo Reply” (type 0, code 0)

# Traceroute e ICMP



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# Traceroute e ICMP

```
C:\Prompt de Comando

C:\Users\willrich>tracert www.ietf.org

Rastreando a rota para www.ietf.org [12.22.58.30]
com no máximo 30 saltos:

 1    16 ms    6 ms    *    150.162.239.253
 2    29 ms    25 ms   16 ms   popsc-10g-ufsc-te-1-2-rt1.bb.pop-sc.rnp.br [200.
237.194.45]
 3    30 ms    71 ms    9 ms   xe-2-0-0-2910-r0-sc.bkb.rnp.br [200.143.254.161]
 4   374 ms   183 ms   117 ms  xe-3-1-1-3000-r0-sp.bkb.rnp.br [200.143.252.65]
 5    158 ms   173 ms   179 ms  200.143.254.234
 6    133 ms   134 ms   177 ms  66.165.175.25
 7     *      168 ms   *      t0-0-0-5.br2.mia.terremark.net [66.165.161.93]
 8     *      213 ms  161 ms  12.88.168.13
 9    210 ms    *      *      cr1.ormfl.ip.att.net [12.122.143.50]
10    247 ms    *      218 ms  cr2.hs1tx.ip.att.net [12.122.1.5]
11    218 ms    *      *      cr1.dlstx.ip.att.net [12.122.28.157]
12    203 ms   206 ms   218 ms  cr2.dlstx.ip.att.net [12.122.1.210]
13    249 ms   223 ms   215 ms  cr2.la2ca.ip.att.net [12.122.28.178]
14    205 ms   206 ms   225 ms  cr1.la2ca.ip.att.net [12.122.2.165]
15    305 ms    *      230 ms  cr82.sj2ca.ip.att.net [12.122.1.146]
16    243 ms   256 ms   *      gar14.sffca.ip.att.net [12.122.110.9]
17     *      205 ms   214 ms  12.94.198.6
18    205 ms   232 ms   220 ms  mail.ietf.org [12.22.58.30]

Rastreamento concluído.

C:\Users\willrich>
```

# Pontos Importantes

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## Protocolo ICMP e Traceroute

- Entender os objetivos do protocolo ICMP
- Conhecer o funcionamento do traceroute